Network Engineering and Design Project

Digital Network Design and Routing Documentation

Applying VLSM and Static Routing in a Two■Tier Architecture

Prepared by: Mohamed Ashraf

Target Certification: Cisco Certified Network Associate (CCNA)
Software Used: Cisco Packet Tracer
Documentation Date: 30 May 2026

1. Project Overview

This project designs and implements a Two■Tier Architecture connecting four LANs. VLSM (Variable Length Subnet Mask) is used to efficiently allocate IP addresses. Static Routing is configured between routers to provide full connectivity between all devices.

2. Network Topology

- Router R1 connects LAN 1 and LAN 2.
- Router R2 connects LAN 3 and LAN 4.
- A WAN Point■to■Point link connects R1 and R2 through Gig0/2.

3. VLSM Address Planning

Base network: 192.168.5.0/24.
Subnets were allocated according to host requirements while avoiding overlapping addresses.

4. Addressing Table

LAN 2 (64 Hosts): 192.168.5.0/25, Gateway 192.168.5.1
LAN 1 (45 Hosts): 192.168.5.128/26, Gateway 192.168.5.129
LAN 3 (14 Hosts): 192.168.5.192/28, Gateway 192.168.5.193
LAN 4 (9 Hosts): 192.168.5.240/28, Gateway 192.168.5.241
WAN Link: 192.168.6.0/30
R1 G0/2: 192.168.6.1
R2 G0/2: 192.168.6.2

5. Static Routing Configuration

Router R1:
ip route 192.168.5.192 255.255.255.240 gigabitEthernet 0/2
ip route 192.168.5.240 255.255.255.240 gigabitEthernet 0/2

Router R2:
Configured with corresponding interfaces and static routes toward remote networks.

6. Verification and Testing

- Verified routing table using 'show ip route'.
- Performed end■to■end ping tests between PCs.
- Connectivity success rate reached 100% with zero packet loss.

Conclusion:
The project successfully demonstrates VLSM subnetting and static routing implementation in a Two■Tier architecture suitable for CCNA practice.